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MASTER THESIS

TITLE: Morphology-Aware Tokenization for Malayalam: Design and Evaluation

GOAL:

To develop and evaluate a morphology-aware hybrid tokenization approach for Malayalam and examine its impact on tokenization quality and downstream NLP task performance.

RESEARCH QUESTIONS:

1. To what extent can a rule-based "morphological shield" reduce fragmentation of agglutinative Malayalam word forms produced by standard BPE tokenization under a fixed vocabulary setting?
2. Under controlled experimental conditions, does morphology-aware hybrid tokenization produce statistically measurable differences in neural topic classification performance compared to standard BPE?
3. Is there a statistically observable relationship between intrinsic tokenization quality metrics (e.g., fragmentation rate, fertility, boundary violation rate) and downstream classification performance in Malayalam?

TASKS:

- **Linguistic Framework Formalization:** Formalize a constrained and computationally feasible set of morphological preservation rules targeting high-frequency agglutinative structures in Malayalam, including suffix stacking, inflectional endings, and selected compound constructions. Implement these rules as a pre-tokenization constraint layer designed to preserve specified boundary patterns without attempting full morphological analysis.

- **Hybrid Tokenizer Design:** Train a standard BPE tokenizer on a curated Malayalam corpus under fixed vocabulary size conditions. Develop a hybrid tokenization pipeline by integrating morphological boundary constraints into the BPE training workflow,

producing two directly comparable vocabularies: standard BPE and morphology-aware hybrid BPE, while controlling for corpus, vocabulary size, and training parameters.

- **Intrinsic Empirical Evaluation:** Compute and compare intrinsic tokenization metrics across both tokenizers, including Word Fragmentation Rate (WFR), average token fertility, morph-boundary violation rate, and token length distribution. Conduct comparative statistical analysis to quantify differences under identical evaluation conditions.
- **Downstream Task Performance Study:** Construct or curate a balanced Malayalam topic classification dataset with clearly defined class distributions. Train a lightweight Transformer-based neural model under identical architectural and hyperparameter configurations using each tokenization scheme (standard BPE and hybrid BPE). Evaluate classification accuracy, macro-F1 score, and convergence stability across multiple controlled runs to ensure statistical robustness.

- **Comparative Synthesis:** Perform a structured comparative analysis examining whether observed intrinsic differences correspond to statistically measurable differences in downstream performance. Analyze potential trade-offs between morphological boundary preservation, vocabulary efficiency, and computational considerations within the defined experimental scope.

EXPECTED RESULTS:

1. **A Controlled Hybrid Tokenization Framework:** A formally specified and implemented morphology-aware BPE pipeline designed for reproducibility and comparative evaluation within Malayalam NLP.
2. **Experimental Artifact:** An open-source Malayalam tokenizer implementation with documented training procedures and evaluation scripts to facilitate replication under similar experimental conditions.
3. **Empirical Comparative Data:** Quantitative and statistically analyzed comparison between standard and hybrid tokenization across intrinsic metrics and downstream topic classification performance.
4. **Validation Findings:** Empirical evaluation of the extent to which controlled improvements in morphological boundary preservation influence neural topic classification performance in Malayalam, supported by statistical analysis of observed effects and explicit discussion of limitations.

<u>WORK PLAN</u>

Period	Activities	Expected results

20-02-2026	Co-ordination of the Project	Selecting the research topic and plan agreed with the supervisor
01-03-2026	Preparation of the Topic Plan	Approving the Topic and Plan to Upload to the Moodle
15-04-2026	Preperation of the Intermediate defence	Intermediate Defence
16-04-2026	Updation and Correction from the Intermediate Defence	Research Project Submission and Research project to be Uploaded to the moodle
20-05-2026	Preperation for the Final Defence of Master Thesis	The Submission of Final Version of the research project.

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Supervisor — Mindaugas Petkevičius

Study Programme Coordinator

(title, first name, last name signature)